

Solving $Ax = b$

Can this target be hit? And if so, what is **every** portfolio that hits it?

The rank alone tells you: none, one, or infinitely many.

$x = x_p + x_n$

Rank cases

MIT 18.06, Lecture 8

THREE OUTCOMES

One, none, or infinitely many

Funds (budget, return)	Target	Outcome
Nifty + bonds	(10, 105)	one: (7, 3)
Nifty + its twin ETF	(10, 120)	infinitely many
Nifty + its twin ETF	(10, 105)	none: $0 = -15$

Never exactly two: if two portfolios work, their difference is a do-nothing trade, and every multiple of it works too.

Four funds, a target (B, R, E)

Nifty, Nifty ETF, bonds, balanced:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} x = \begin{bmatrix} B \\ R \\ E \end{bmatrix}$$

Eliminate with the target carried along as letters.

A zero row becomes a condition

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & B \\ 0 & 0 & -5 & -2.5 & R - 12B \\ 0 & 0 & 0 & 0 & \frac{1}{5}(7B + 5E - R) \end{array} \right]$$

Row 3 says $0 = \frac{1}{5}(7B + 5E - R)$. So a solution exists exactly when

$$R = 7B + 5E$$

The same plane $C(A)$ as Lesson 5: algebra and picture agree.

TEST IT

Which targets pass?

Target	$7B + 5E - R$	
(10, 105, 7)	$70 + 35 - 105 = 0$	✓
(10, 120, 10)	$70 + 50 - 120 = 0$	✓
(10, 105, 5)	$70 + 25 - 105 = -10$	✗

Each vanished row gives one condition. Meeting them all is the same as " b is in $C(A)$ ".

The particular solution

Set the free amounts (ETF, balanced) to 0 and back-substitute for $b = (10, 105, 7)$:

$$-5x_3 = 105 - 120 = -15 \Rightarrow x_3 = 3$$

$$x_1 + 3 = 10 \Rightarrow x_1 = 7$$

so

$x_p = (7, 0, 3, 0)$: ₹7 lakh Nifty, ₹3 lakh bonds.

ALL ANSWERS

$$\mathbf{X} = \mathbf{X}_p + \mathbf{X}_n$$

$$x = \begin{bmatrix} 7 \\ 0 \\ 3 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix}$$

$c = d = 2$: $(4, 2, 2, 2)$. Budget 10 ✓, return $48 + 24 + 14 + 19 = 105$ ✓, exposure 7 ✓.

One portfolio + any do-nothing trade = every portfolio.

Read it off $[R \mid d]$

$$[R \mid d] = \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0.5 & 7 \\ 0 & 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Free amounts 0 $\Rightarrow$ pivot amounts = d : $x_1 = 7$, $x_3 = 3$. The free columns give the special solutions. One matrix, both halves of the answer.

Four cases

Rank	Solutions	Funds
$r = m = n$	exactly 1	Nifty, bonds, midcap
$r = n < m$	0 or 1	Nifty, bonds, 3 rules
$r = m < n$	∞ , always	Nifty, bonds, midcap, balanced
$r < m, r < n$	0 or ∞	Nifty, ETF, bonds, balanced

$r = m$: always solvable. $r = n$: never more than one.

Every way to build a payoff

Futures, 20,000 call, 20,000 put, 19,900 call: rank 3 = 3 scenarios, so every payoff is buildable. For $(-200, 100, 900)$:

$$x = (1, 2, 0, 1) + t(1, -1, 1, 0)$$

"1 fut + 2 calls + 1 $\times$ 19,900 C" = "3 fut + 2 puts + 1 $\times$ 19,900 C". Quote whichever is cheaper.

Key takeaways

1. One, none or infinitely many: never two.
2. Zero rows of $[U \mid c]$ give the conditions on b .
3. x_p : free amounts = 0.
4. All solutions: $x_p + x_n$.
5. $[R \mid d]$ gives x_p directly.
6. $r = m \Rightarrow$ solvable; $r = n \Rightarrow$ unique.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A target tester, a complete-solution lab for any matrix up to 3×5 , and worked hedging and options examples:

learn.krisarj.com

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